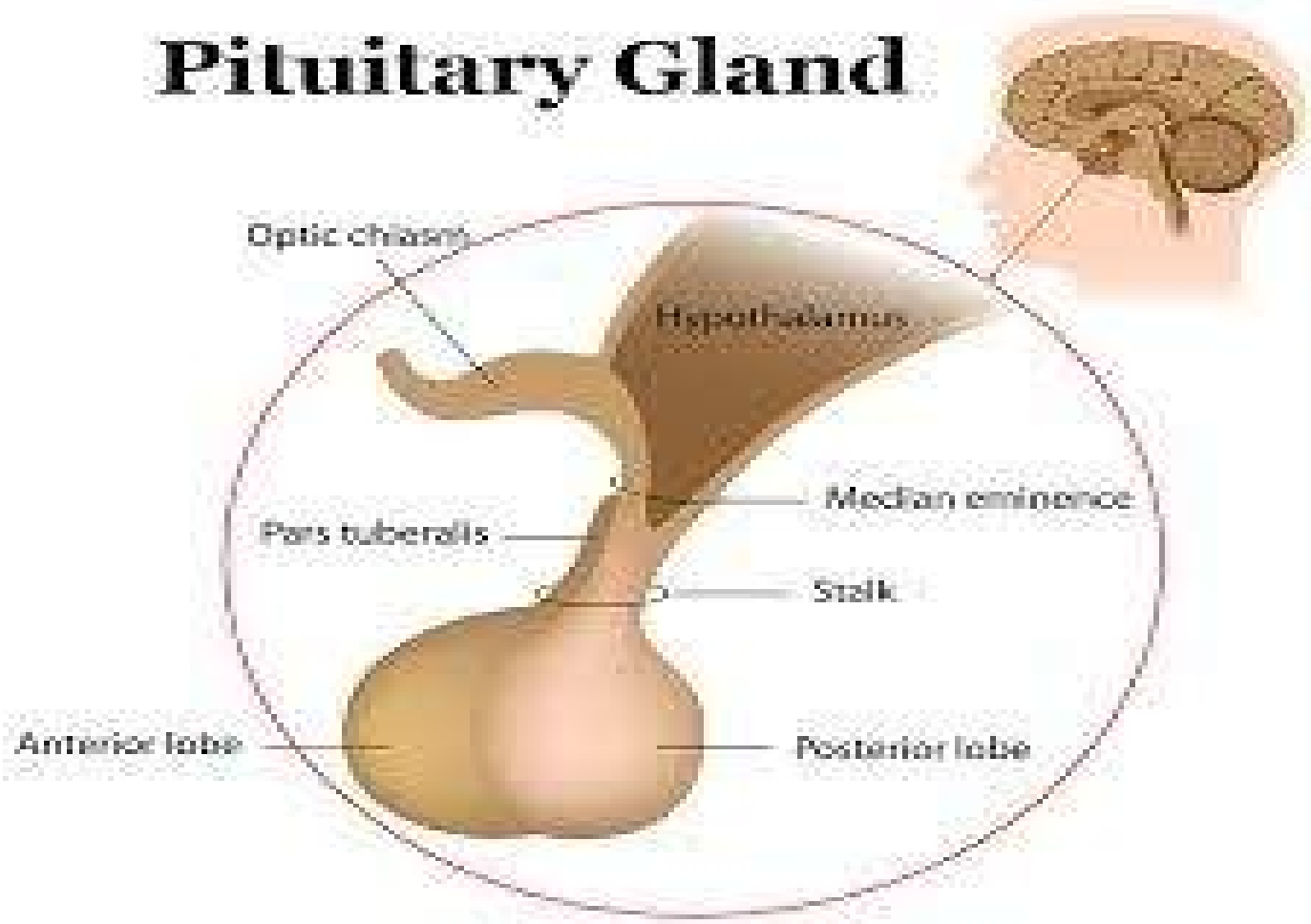
Pituitary Gland

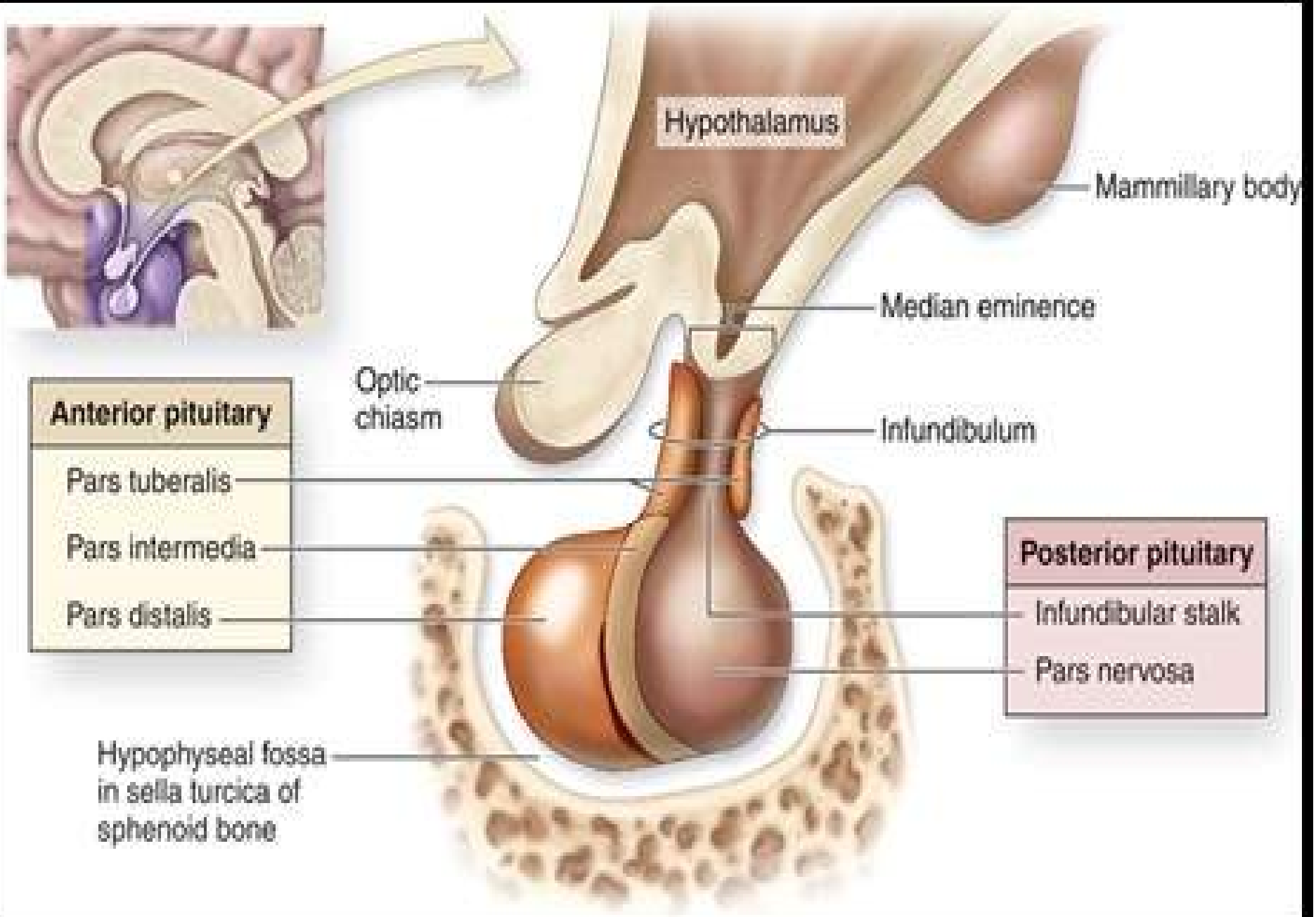


OBJECTIVES

- At the end of the lecture, student would be able to:
  - Describe the **anatomy of pituitary gland**.
  - Know the **hormones of anterior and posterior pituitary gland.**
  - Explain the **physiology** of different hormones.

PITUITARY GLAND

- Syn: **Hypophysis or Hypophysis Cerebri**
- Anatomy:
  - Lies: with in the Sella Turcica at the base of the brain and is connected to the Hypothalamus by means of Pituitary stalk or infundibulum
  - Average weight: 0.5 gm
  - Divided into: Anterior pituitary (Adenohypophysis) and Posterior pituitary (Neurohypophysis)

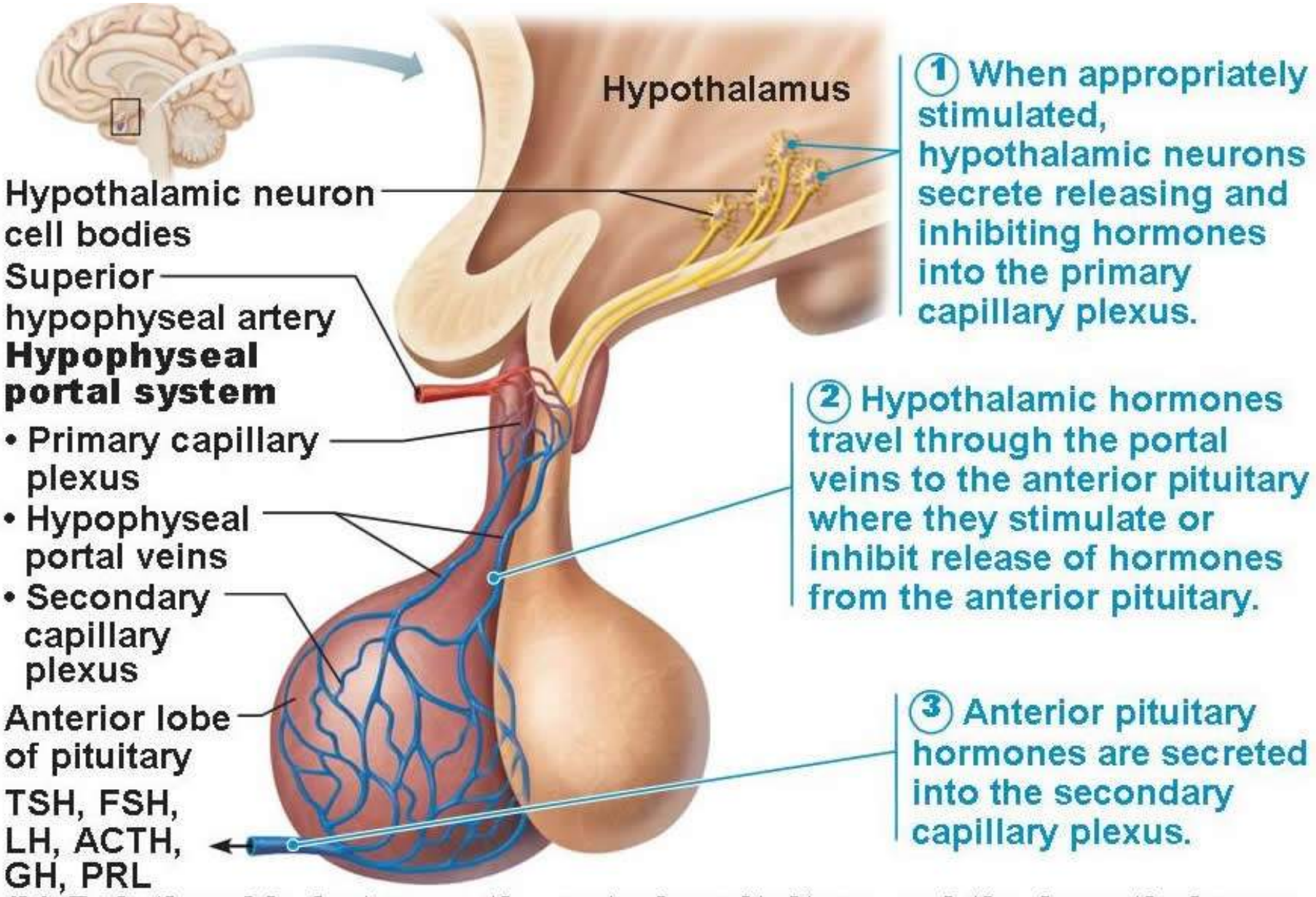

Hypothalamus
Mammillary body
Median eminence
Optic chiasm
Infundibulum
Anterior pituitary
Pars tuberalis
Pars intermedia
Pars distalis
Posterior pituitary
Infundibular stalk
Pars nervosa
Hypophyseal fossa in sella turcica of sphenoid bone

Cont..

- Although closely related anatomically, they are embryologically and functionally quite distinct.
  - The ant. Pituitary comprises primarily glandular tissue
  - While the post pituitary is of neural origin.

BLOOD SUPPLY OF ADENOHYPOPHYSIS

- **The ant. Pituitary has one of the highest blood flow of any tissue in the body. Its blood supply is portal in character k/a HYPOTHALAMIC-HYPOPHYSEAL PORTAL SYSTEM.**
- **FUNCTION: This portal system carries hypothalamic releasing and inhibitory hormones from median eminence to ant. Pituitary gland.**



(b) Relationship between the anterior pituitary and the hypothalamus

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HORMONES OF ADENOHYPOPHYSIS

(in past called the master gland)

- **It secretes at least 6 hormones:**

1. **The Growth hormone (GH) or Somatotropic or somatotropin**
2. **The Thyroid stimulating hormone (TSH) or Thyrotropin**
3. **The Adrenocorticotropic hormone (ACTH) or corticotropin**
4. **The follicle stimulating hormone (FSH)**
5. **The Luteinizing hormone (LH)**
6. **Prolactin or lactogenic or lactotropin**

Gonadotropic

- **In addition it also releases beta-lipotropin, MSH and beta-endorphin**

Hormones of anterior pituitary

Acro-nym	Full name	Chemical nature	Mol.wt. in kD	Amino acids
GH	Growth hormone	Polypeptide	22	191
ACTH	Adrenocor-ticotropic hormone	Polypeptide	4.5	39
LH	Luteinizing hormone	Glycoprotein; α,β chains	29	α = 89 β= 115
FSH	Follicle stimulating hormone	Glycoprotein; α,β chains	29	α = 89 β= 115
TSH	Thyroid stimulating hormone	Glycoprotein; α,β chains	28	α = 89 β= 114
α MSH	Melanocyte stimulating hormone	Polypeptide	13	
PRL	Prolactin	Polypeptide	22	198
	β Endorphins	Polypeptides	4	31

HISTOLOGY

- Adenohypophysis consists of 2 types of cells
  - **CHROMOPHOBE CELLS:** Parent cells, no granules stain lightly
  - **CHROMOPHIL CELLS:** large number of granules and actively secrete hormones. These cells are subdivided into two groups:
    - **ACIDOPHIL OR ALPHA CELLS**
    - **BASOPHIL OR BETA CELLS**

- The acidophil cells; 2 types
  - **SOMATOTROPES (50%):** secrete somatotropin
  - **MAMMOTROPES (20%):** secrete prolactin
- The basophil cells: 3 types
  - **CORTICOTROPES 15%**
  - **THYROTROPES 5%**
  - **GONADOTROPES 10%**

REGULATION OF THE SECRETION OF ADENOHYPOPHYSIS

1. **Negative feedback system**
2. **Positive feedback system**
3. **The Hypothalamus: thru hypothalamo-hypophyseal portal vessels:**

The secretion of hormones by adenohypophysis or anterior pituitary is under the control of peptides secreted by hypothalamus

1. **Growth hormone releasing hormone(GHRH)**
2. **Growth hormone inhibitory hormone (GHIH) or Somatostatin**

3. **Thyrotropin releasing hormone (TRH)**
4. **Prolactin inhibitory factor (PIF)**
5. **Corticotropin releasing factor (CRF)**
6. **Luteinizing hormone releasing hormone (LHRH) or Gonadotropin releasing hormone (GnRH): causes release of both LH and FSH.**

Major hypothalamic factors

Name	Chemical nature	Biological actions
TRH; thyrotropin releasing hormone	Tripeptide; (pyro-Glu-His-Pro-NH_2)	Induces secretion of TSH and PRL; neuromodulator
GnRH; gonadotropin releasing hormone	Biologically active portion is a decapeptide	Releases LH and FSH; induces spermatogenesis, ovulation and testosterone
GHRH; growth hormone releasing hormone	37-44 amino acid; amino terminal end is tyrosine	Stimulates growth hormone secretion
CRF; corticotropin releasing factor	Amidated peptide with 41 amino acids	Release of ACTH. Inhibited by cortisol.
Somatostatin; growth hormone inhibitory factor	Cyclic peptide with 14 amino acids	Inhibits secretion of GH and TSH. Inhibits gut hormones, pancreatic and gastric secretion
PIF; prolactin inhibitory factor	Dopamine	Inhibits PRL release

Cont..

- Some examples of the release of ant. Pituitary hormone in response to external stimuli which act through the CNS are given below:

I. Cold increases the secretion of TSH

II. Stress increase the secretion of ACTH

III. Suckling brings about a secretion of oxytocin.

It is now established that the hypothalamus is the highest integrative centre for both the autonomic and endocrine system, endocrine part produces neurohormones of two types 1. releasing 2. release inhibiting hormones.

HORMONES OF THE ADENOHYPOPHYSIS

- **1. THE GROWTH HORMONE (GH):**
  - **It is a single chain of 191 amino acids**
  - **A protein with the same amino acid sequence as human GH has been synthesized and is used in patients**
  - **Secreted by somatotropes**
  - **Maximum level of GH is seen during deep sleep.**
  - **Secretion is low in infancy, increases slightly in early childhood and greatly increases during puberty. After the third decade there is a progressive decrease in its secretion.**

ACTIONS OF GH

- **ON PROTEIN METABOLISM (ANABOLIC):** Growth hormone brings about positive nitrogen balance by retaining nitrogen.
  - Inc. amino acid entry into cells like insulin
  - In. protein synthesis by ribosomes
  - Inc. transcription of DNA to form mRNA
  - Dec. protein catabolism b/c GH mobilizes free FA to supply energy
  - Dec. gluconeogenesis from proteins
  - Collagen synthesis
  - Inc,. Ab formation

ON FAT METABOLISM (LIPOLYTIC):

- Mobilizes free fatty acids from adipose tissues
- Inc. free fatty acids conc. in body fluids
- Inc. conversion of free fatty acids into acetyl CoA for use of energy
- Ketosis: due to inc. acetoacetic acid formation by liver
- Fatty liver: due to excess mobilization of free fatty acid from adipose tissue

- **ON CARBOHYDRATE METABOLISM: (Hyperglycemic)**

  Anti- insulin properties, diabetogenic effect

  - Dec. utilization of glucose by glycolysis due to –ve feedback inhibition by excess acetyl CoA
  - Inc. intestinal absorption of glucose, poor sugar tolerance and glycosuria
  - Inc. glycogen deposition
  - Dec. glucose entry into cells
  - It reduces insulin sensitivity and thereby decreases the hypoglycaemic effect of insulin

- **ON CARTILAGE & BONE:**

  GH acts on liver, somatomedins C (small proteins also known as insulin-like growth factor 1 or IGF-1) formed by liver, act on cartilage and bone to stimulate their growth.

- **ION or MINERAL METABOLISM:**

  It is observed that the intestinal absorption of calcium is increased by GH, since the bone growth and development is stimulated by growth hormone.

  Growth hormone retains Na, Ca, K, Mg and PO_4.

- **ON SOFT TISSUES:**
- Deposition of connective tissue
- Thickening of skin
- Inc. growth of viscera eg liver kidney
- Inc. milk secretion due to prolactin like action

- **ON GROWTH:**
- GH promotes growth of all tissues due to inc. size of cell and inc. mitosis

REGULATION OF GH

- **STIMULATORY FACTOR:**
  - Acute hypoglycemia
  - Chronic hypoproteinemia
  - Dec. plasma FFA
  - Stress, trauma, emotion, exercise
  - Starvation
  - GHRH
  - Catecholamines

- **INHIBITORY FACTORS:**
  - GHIH
  - Glucagon
  - Hyperglycemia
  - Hyperlipidemia
  - Hypercholesteremia

THYROID STIMULATING HORMONE

- K/as THYROTROPIC or THYROTROPIN.
- The α-subunit of TSH, LH, HCG and FSH are nearly identical.
- The biological specificity of thyrotropin must therefore be in β-subunit.
- Metabolic role:
  - There are glycoprotein receptors on the thyroid cell membrane which bind to the receptor binding site on β-subunit of TSH.

cont..

–The TSH stimulates the synthesis of thyroid hormones at all stages such as Iodine uptake by the thyroid cells resulting in a raised ratio thyroid iodine/serum iodine (350:1)

–It enhances the release of stored thyroid hormones by increasing the proteolysis of thyroglobulin

–Increases the number of thyroid cells whose size and secretory activity is also increased.

REGULATION OF TSH

- After action, TSH is rapidly inactivated mainly by the thyroid gland itself.

- TSH secretion is regulated by the plasma level of thyroid hormone and by the hypothalamic thyrotropin releasing hormone.

THE ADRENOCORTICOTROPIC HORMONE

- ACTH/ Corticotropin with mol. wt. 4,500
- Consist of a 39 amino acids, biological activity of ACTH resides in the first 23 amino acids from N-terminal end.
- ACTH has been synthesized
- Its biosynthesis occurs from a large precursor molecule called pro-opio-melano-cortin (POMC).
- Various proteolytic enzymes hydrolyse POMC to give different peptides. Thus POMC is broken down into ACTH, β-lipotropin (LPH). β-LPH is further cleaved into γ-LPH and endorphins.

ACTIONS OF ACTH

- ✓ **ADRENAL EFFECTS:**
- Stimulates zona fasciculata and zona reticularis of the adrenal cortex to secrete cortisol and androgens.
- Stimulates zona glomerulosa to secrete aldosterone(much smaller than on other hormones).
- Patients with a decreased ACTH production have near normal aldosterone production.
- After hypophysectomy , the zona fasciculata and zona reticularis become markedly thinner.

Cont..

- Following changes are seen to take place in the adrenal cortex when ACTH is administered
  - The size of the adrenal cortex increases within a few hours; the adrenal blood flow is increased
  - There is an increased glycogenolysis, increased protein synthesis and increased cAMP contents
  - Increased synthesis of corticosteroids
  - The cholesterol content of the adrenal cortex falls, because it is converted to steroid hormones

REGULATION of ACTH

1. THE HYPOTHALAMUS: Produces corticotropin releasing hormone(CRH), greatly increases the synthesis and release of ACTH.

- Under resting conditions, the rate of secretion of CRH is low, but in stressful conditions e.g. hypoglycemia, hypovolemia, hypotension, irradiation with x-rays, trauma, surgical operations , infections, exposure to excessive cold or heat, debilitating disease and emotional excitement , the secretion of CRH is greatly increased.
- Plasma ACTH level is the highest just prior to awaking and is lowest shortly before sleeping k/a SLEEP-WAKE DIURNAL RHYTHM.

2. PLASMA CORTISOL LEVEL:

The level of the free cortisol in the plasma acts by a negative feedback action. Raised level decreases while a low plasma cortisol level stimulates ACTH.

3. OTHER FACTORS: An increased ACTH release is brought about by vasopressin, Interleukin-I and naloxone.

DEFECTIVE RESPONSE OF ADRENAL CORTEX TO ACTH:

1. Atrophy of the adrenal cortex in hypopituitarism or after hypophysectomy.

2. Administration of cortisol in high doses over a prolonged period. This result in an atrophy of the adrenal cortex by inhibiting the secretion of ACTH. Abrupt stoppage results in ACUTE ADRENOCORTICAL INSUFFICIENCY or CRISIS.

✓ EXTRA-ADRENAL EFFECTS OF ACTH:

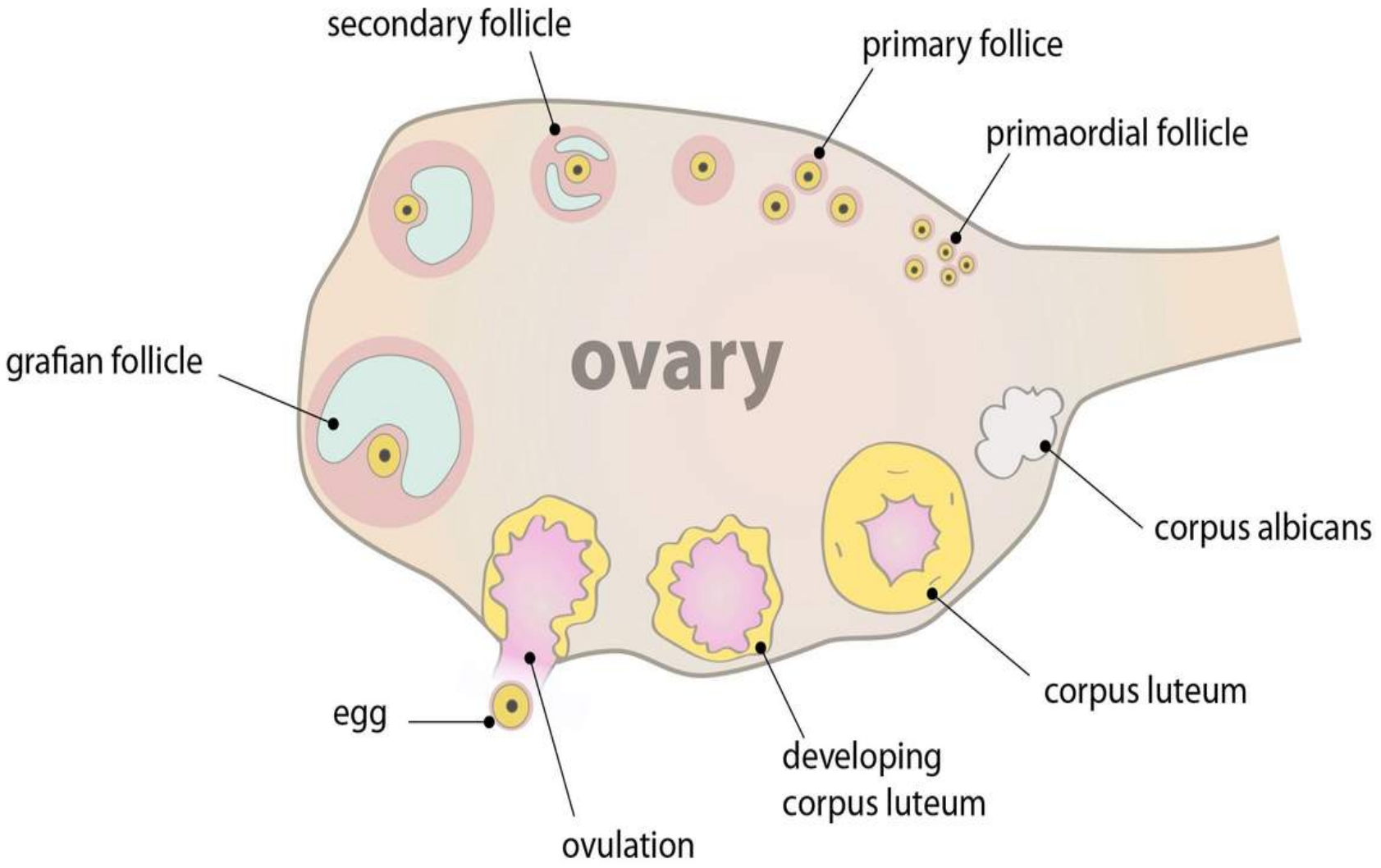
- Release of free fatty acids from fat depots
- Slight diabetogenic action
- Release of insulin results in lowering of blood glucose
- Decreased deamination of amino acids
- Deposition of glycogen in adipose tissue is increased, regarded as due to stimulation of insulin secretion
- Pigmentation of the skin

GONADOTROPIC HORMONES

- **Two Gonadotropins :**
  - **FOLLICLE STIMULATING HORMONE (FSH),**
  - **LUTEINIZINGHORMONE (LH) ; In male LH is called Interstitial cell stimulating hormone (ICSH)**
- **Glycoprotein in nature.**
- **Exert their effects on gonads, thus vital role in reproduction.**

Physiological actions in female

- Marked increase in size and weight of the ovaries
- Promotes follicular growth and maturation of ovarian follicle
- Ovulation d/t LH
- Corpus luteum formation


secondary follicle
primary follice
primaordial follicle
grafian follicle
ovary
corpus albicans
corpus luteum
egg
developing
corpus luteum
ovulation

• HORMONAL REGULATION OF OOGENSIS AND OVULATION

OVULATION

FOLLICULAR PHASE

10-20 primordial follicles begin to develop in response to FSH and LH levels

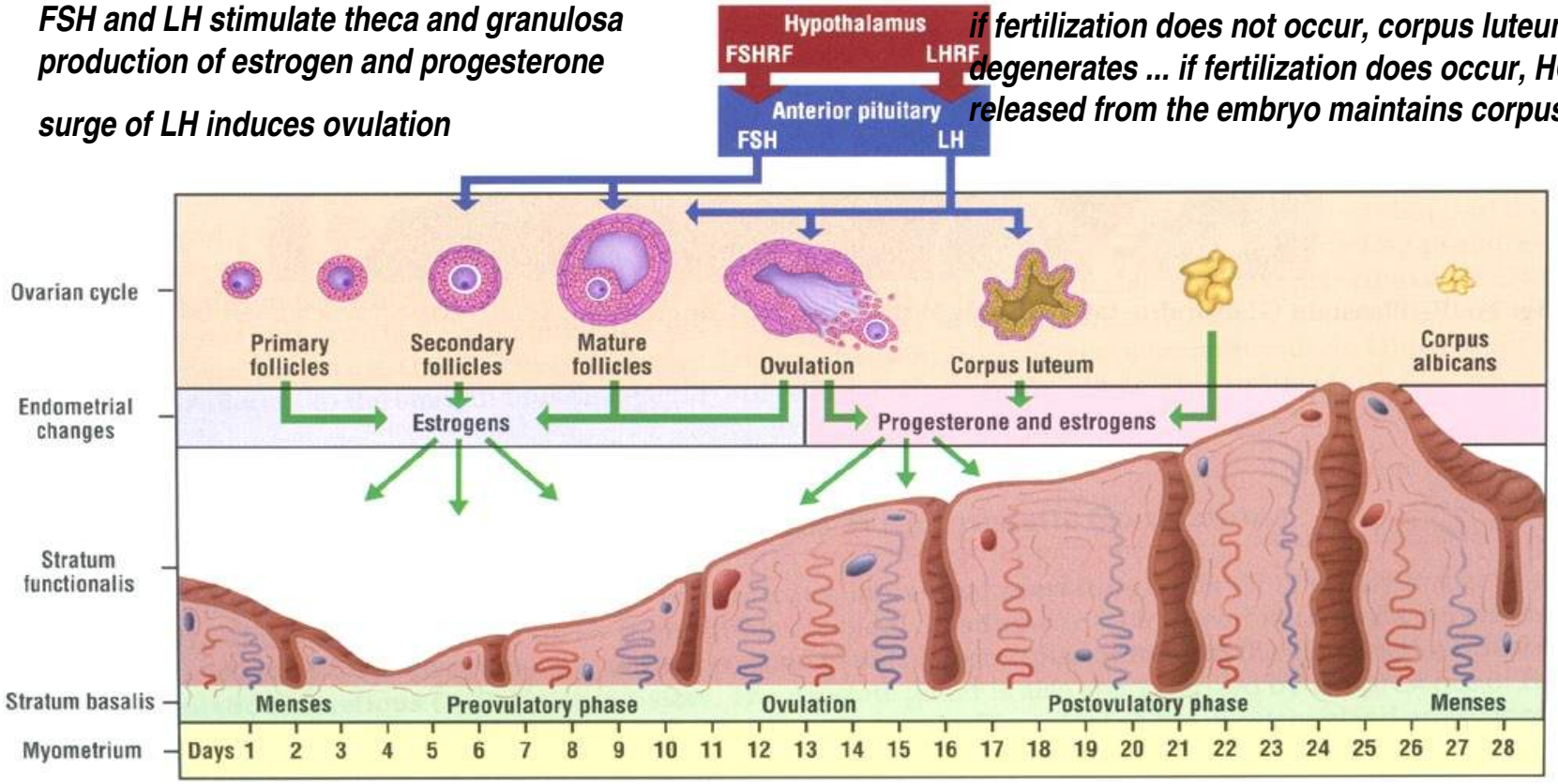
FSH and LH stimulate theca and granulosa production of estrogen and progesterone

surge of LH induces ovulation

LUTEAL PHASE

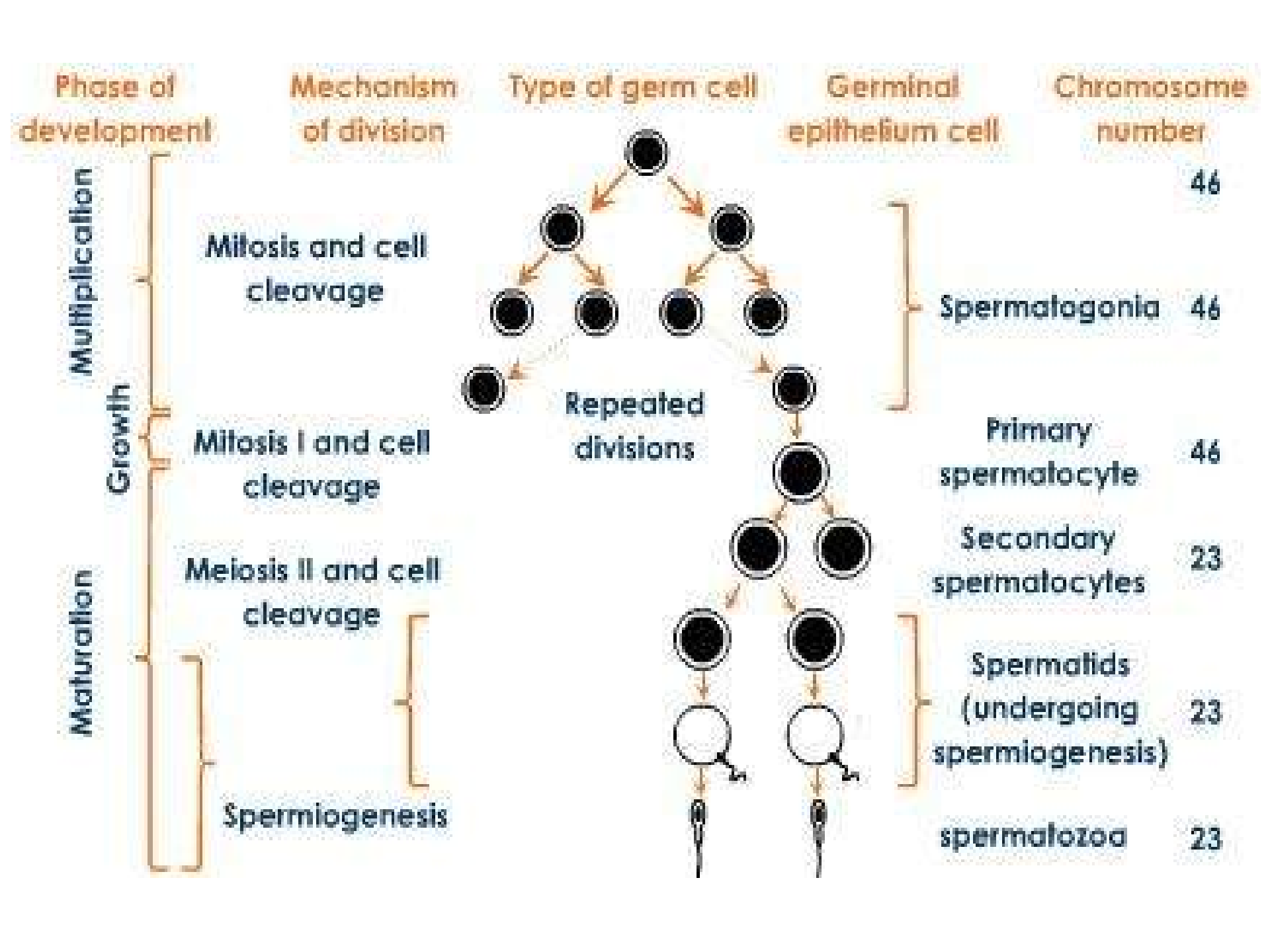
theca and granulosa cells transform into the corpus luteum and secrete large amounts of progesterone

if fertilization does not occur, corpus luteum degenerates ... if fertilization does occur, HCG released from the embryo maintains corpus luteum



Physiological actions in male

- Regulation of two main functions:
  - Spermatogenesis
  - Formation of testosterone, required for last step in spermatogenesis
- LH causes the secretion of testosterone from Leydig cells.


Phase of development
Mechanism of division
Type of germ cell
Germinal epithelium cell
Chromosome number
Multiplication
Growth
Maturation
Mitosis and cell cleavage
Mitosis I and cell cleavage
Meiosis II and cell cleavage
Spermiogenesis
Repeated divisions
Spermatogonia
46
Primary spermatocyte
46
Secondary spermatocytes
23
Spermatids (undergoing spermiogenesis)
23
spermatozoa
23

REGULATION OF GONADOTROPIN

- **CENTRAL NERVOUS SYSTEM:**
  - Gonadotropin releasing hormone (GnRH),
  - Psychological factors e.g, fear, emotional upset etc lead to abnormalities of mensturation
- **ROLE OF PLASMA SEX HORMONE LEVEL:**
  - Raised plasma estrogen inhibit FSH and GnRH.

Cont..

- **LH SURGE: (double role of estrogen)**
  - In follicular phase, plasma estrogen level rises and inhibits secretion of LH
  - 36-48hrs before ovulation there is pulsatile increase in plasma estrogen that brings about positive feedback and causing increased secretion of LH, followed by ovulation.

PROLACTIN

- Mammotropic hormone/Lactogenic hormone/lactotropin
- A protein of 198 amino acids
- It mostly affects the growth of the mammary gland and the secretion of milk
- No physiological role in the male
- Regulated by hypothalamus thru prolactin release inhibitory hormone
- Pregnancy, suckling reflex, TRH, sleep, stress and tranquilizing drugs increases prolactin secretion

- **Hyperprolactinemia in males:**

  Excessive secretion of prolactin is associated with pathological effects e.g. oligospermia, infertility and galactorrhea.

- **Hyperprolactinemia in females:**

  Inhibition of release of GnRH resulting in a decreased FSH and LH, results in hypogonadism, amenorrhea and infertility.

NEUROHYPOPHYSIS

- INTRODUCTION: Histologically it has following structural component:
  - A large number of unmyelinated nerve fibers
  - Numerous blood vessels

- Nerve supply comes from hypothalamus via hypothalamo-hypophyseal tract. Fibers terminate in close proximity to blood vessels.

- Posterior pituitary hormones are synthesized in neurosecretory neurons. They are stored in the pituitary in association with two proteins neurophysin I and II.

- The release of these two hormones is independent of each other.

HORMONES

- Two hormones:
  - VASOPRESSIN/ ANTIDIURETIC HORMOME, ADH
  - OXYTOCIN

Both hormones are nanopeptides, 7 of the 9 amino acids forming each of these hormones are the same.

Oxytocin differs from Vasopressin with respect to 3rd and 8th amino acid residues.

ADH

- Site of formation: Supra-optic nuclei of hypothalamus

- Secretion: by nerve endings in post. Pituitary gland

- Actions:
  - Antidiuretic action: increased water absorption from lumen of collecting tubules and ducts of kindneys, leads to formation of hypertonic urine, conserves water in body.
  - It is released due to rise in plasma osmolarity.

Cont..

- On Vascular Smooth muscle (Pressor effect): It stimulates the contraction of smooth muscles and thus causes vasoconstriction by increasing cytosolic Ca+2 concentration thus causes increased in arterial pressure

- Other actions: contraction of smooth muscle in GIT, bile duct and uterus

- **REGULATION**:
- **Stimulating factors:**
  - Inc. Na+ conc. Or inc. osmolality of ECF
  - Low blood volume and low BP
  - Trauma, pain anxiety
  - Drugs eg, morphine, nicotine, tranquilizers and anesthetics
- **Inhibitory factors:**
  - Dec. Na+ conc or dec. osmolality of ECF
  - Inc. blood vol and inc. BP
  - Alcohol (hence causes diuresis)

Metabolic Role of Oxytocin

- Contraction of smooth muscle is the primary function of oxytocin.
  1. Galactobolic effect: oxytocin stimulates contraction of the myoepithelial cells that surround the mammary alveoli, causing the milk to be ejected into the ducts leading to the nipples.

  2. Uterine effect: It is found to be elevated at full term pregnancy. It causes contraction of uterine muscle for child-birth.

PITUTARY DISORDERS

Anterior Pituitary Disorders

ANTERIOR PITUITARY HYPERFUNCTION

- **PROLACTINOMA with HYPERPROLACTINEMIA:**

- This is the most common pituitary tumor (30%)
- In women, prolactinoma results in amenorrhea and galactorrhea

- **SOMATOTROPIC ADENOMA:**

- Second most common pituitary tumor
- Acidophilic
- Hyperproduction of somatomedins by the liver

- **Gigantism** if adenoma develops before epiphyseal closure

- **Acromegaly** if adenoma develops after epiphyseal closure, characterized by overgrowth of the jaws, face, hands and feet, generalized enlargement of viscera, along with hyperglycemia, osteoporosis, and hypertension.

- Other results include local compression effects due to expansion of the tumor within the sella turcica.




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- **CORTICOTROPIC ADENOMA:**
  - Hypersecretion of ACTH
  - Results in hypercorticism called Cushing disease or Cushing syndrome

- **CUSHING DISEASE:**
  - This condition classically refers to hypercorticism due to a corticotropic adenoma of the pituitary most often a basophilic adenoma

- **CUSHING SYNDROME:**
  - This condition refers to hypercorticism regardless of cause. It is most often of pituitary and less often of adrenal origin; use terms adrenal Cushing syndrome
  - The cause may be ectopic ACTH production by various tumors.

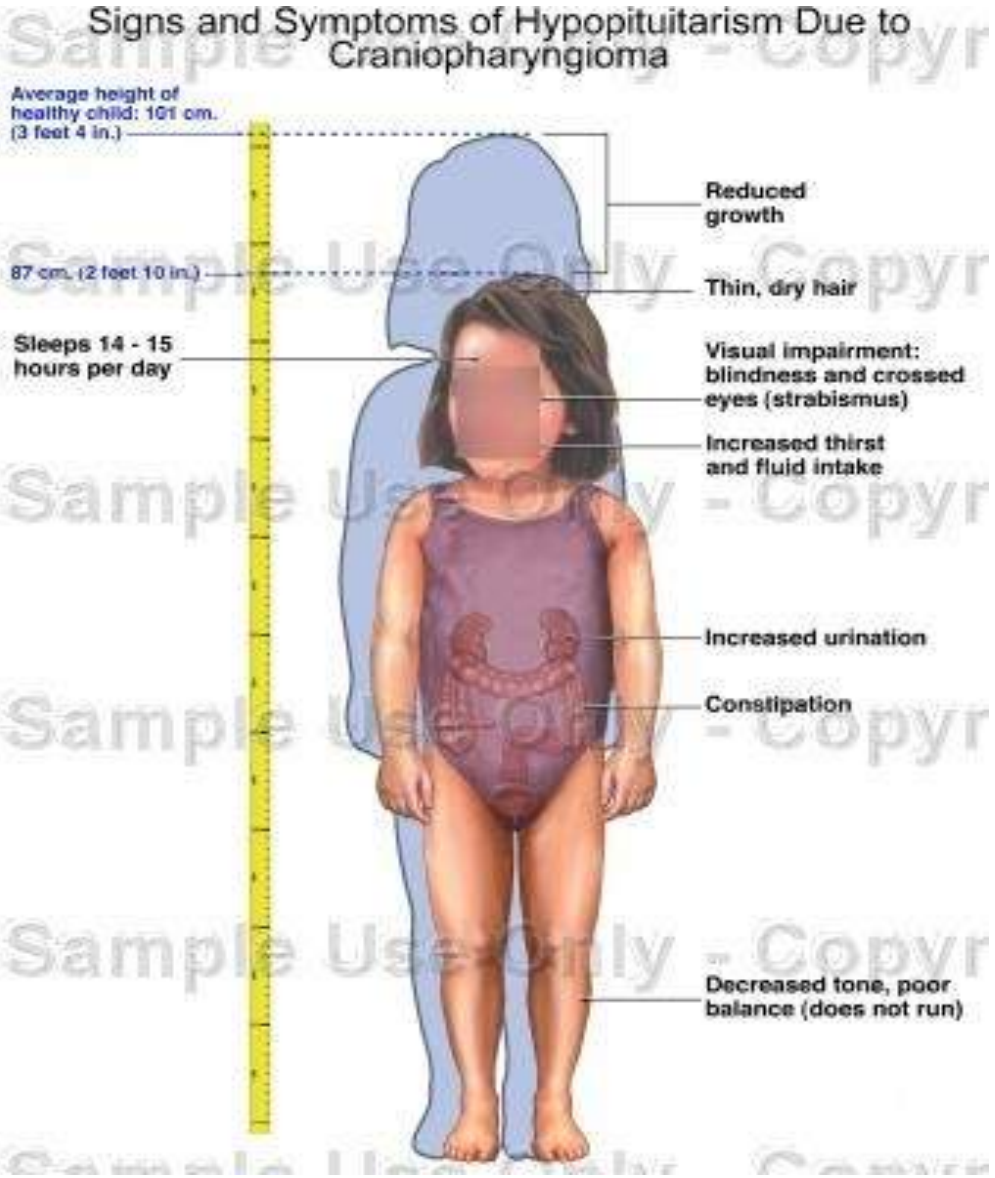
PITUITARY HYPOFUNCTION

- **Pituitary cachexia (Simmonds disease)**
  - Generalized panhypopituitarism
  - Characteristic include marked wasting
- Most frequently caused by:
  - Pituitary tumors
  - Postpartum pituitary necrosis (Sheehan syndrome): caused by ischemic necrosis of the PG and is characteristically associated with hemorrhage and shock during childbirth.
  - Clinical manifestations are due at first to loss of gonadotropins, then to subsequent loss of TSH and ACTH.

Selective deficiency

- **Deficiency of GH:**
  - In children, this results in growth retardation (pituitary dwarfism)
  - In adults, this may result in increased insulin sensitivity with hypoglycemia, decreased muscle strength and anemia.

Decreased GH in child: Dwarfism



Cont..

- Deficiency of gonadotropins:
  - In preadolescent children, this results in retarded sexual maturation.
  - In adults, this results in loss of libido, impotence, loss of muscular mass and decreased facial hair in men, and in amenorrhea and vaginal atrophy in women

Cont..

- Deficiency of TSH results in secondary hypothyroidism.

- Deficiency of ACTH results in secondary adrenal failure.

Posterior Pituitary Disorders:

Deficiency or excess of ADH.

- Diabetes insipidus
- SIADH

Diabetes Insipidus (DI)

- DI is usually insidious but can occur with damage to the hypothalamus or the pituitary (neurogenic DI).
- May be a result of defect in renal tubules, do not respond to ADH (nephrogenic DI).
- Decreased production and secretion of ADH results in massive water loss.
- Leads to hypovolemic & dehydration.

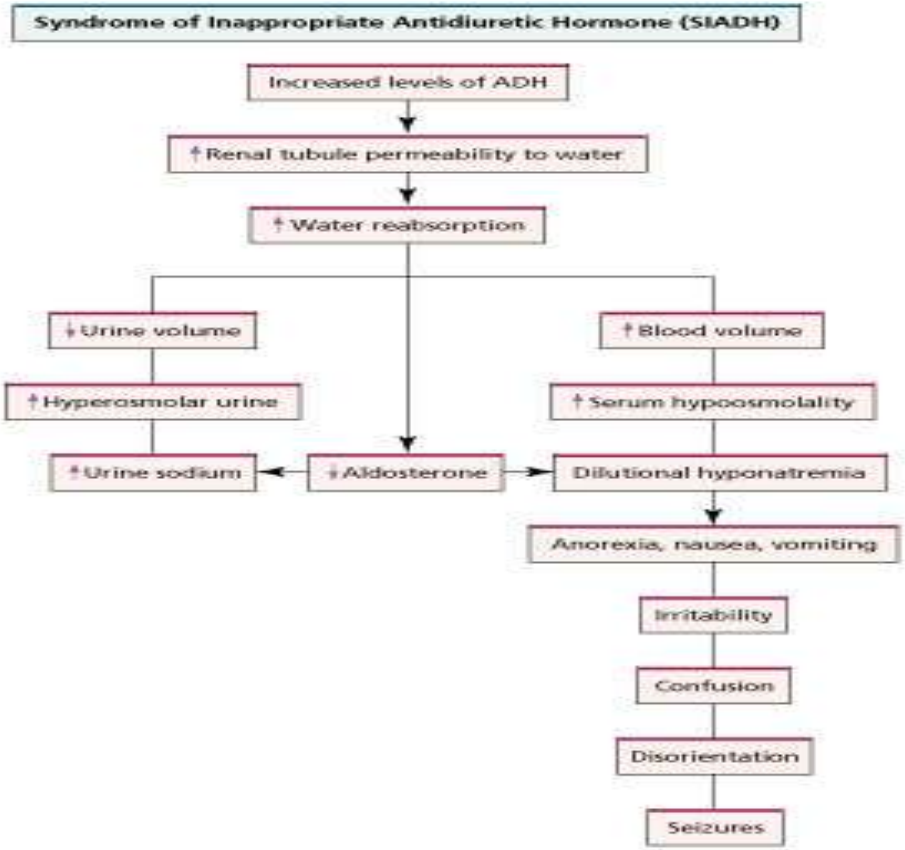
Clinical Manifestations

- DI is characterized by the production of large volumes of dilute urine (polyuria more than 3 litres per day, may be up to 20L).
- Polydipsia (constant thirst)
- Urine specific gravity low
- Weight loss
- Dry skin & mucous membranes
- Possible hypovolemia, hypotension, electrolyte imbalance

SIADH

- Syndrome of Inappropriate Anti-Diuretic Hormone
- Too much ADH produced or secreted.
- SIADH commonly results from excessive production of ADH secreted from posterior pituitary or an ectopic source.
- Often the result of a carcinoma (i.e. lung cell ca), also infections, trauma, drugs.
- Key features are water retention, dilutional hyponatremia and hypo-osmolality
- Decreased serum osmolality cause water to move into cells.

Pathophysiology of SIADH



- Water intoxication, cerebral edema, severe hyponatremia cause altered neurological status, which untreated may cause death.

THANK YOU